

Digital Logic

Design (EE1005)

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Attempt all the questions in the given sequence.

CLO # 2: Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronics circuits, including Boolean Algebra and

Multi-variable Karnaugh map methods.

Design and analysis of digital electronics circuits.

[6 + 2 = 8 Marks]

Sessional-II Exam

Time: 20 Minutes

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 2

Q1: (a) Consider a 3-bit unsigned binary number $X = X_2X_1X_0$. A logical circuit produces an output $Z = Z_2Z_1Z_0$

such that:
 $Z = X - 1$

For example:

- If $X = 000$, then $Z = 111$
- If $X = 101$, then $Z = 100$
- ▶ $Z = 000 + (2\text{'s complement of } 1)$
- ▶ $Z = 101 + (2\text{'s complement of } 1)$
- ▶ $Z = 000 + 111$
- ▶ $Z = 101 + 111$

(i) Construct the complete truth table for the above 3-bit logical circuit.

x_2	x_1	x_0	z_2	z_1	z_0
0	0	0	1	1	1
0	0	1	0	0	0
0	1	0	0	0	1
0	1	1	0	1	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0

(ii) Derive the simplified Boolean expressions for each output bit Z_2, Z_1, Z_0 using K-map minimization.

$x_2 \backslash x_1$	00	01	11	10
0	1	0	1	0
1	0	0	1	1

$$Z_2 = X_2X_0 + X_2X_1 + X_2'X_1'X_0'$$

$x_2 \backslash x_1$	00	01	11	10
0	1	0	0	1
1	0	1	1	0

$$Z_1 = X_1'X_0' + X_1X_0$$

$x_2 \backslash x_1$	00	01	11	10
0	1	1	1	1
1	0	0	0	0

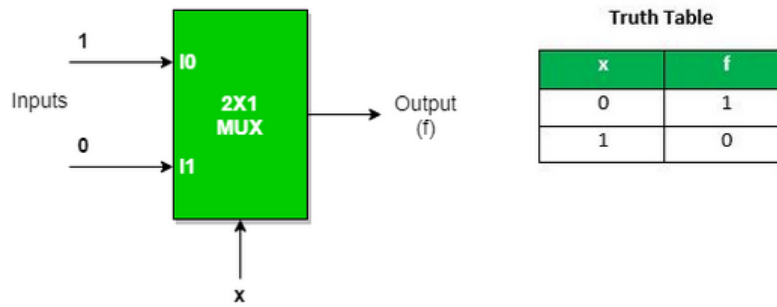
$$Z_0 = X_0'$$

(iii) Draw the logical circuit of the above Boolean expression for Z_2 only using NAND gate.

(b) Implement the NOT gate using a 2:1 Multiplexer.

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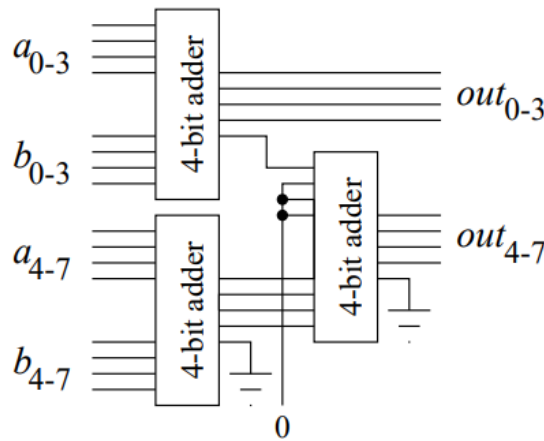
CLO # 3 Analyze small –scale combinational digital circuits.

Design and analysis of Combinational Logic Circuits:

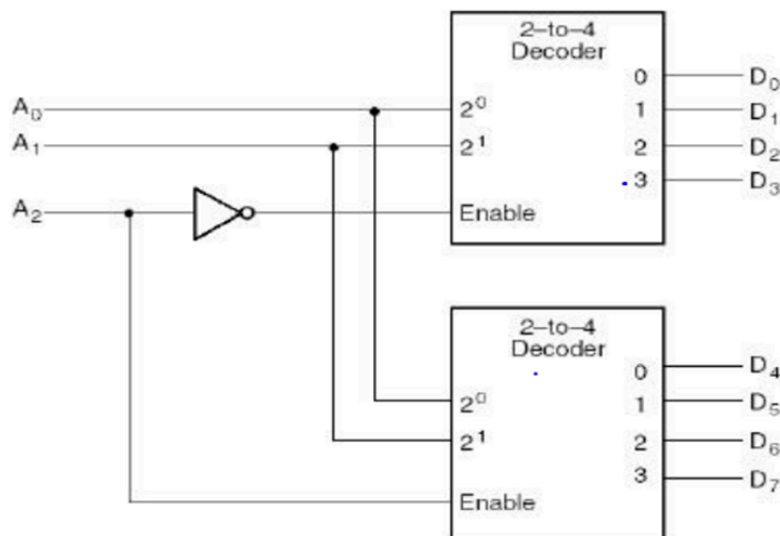
[4 + 4 + 4 + 2 + 2 + 6 = 22 Marks]

Time: 40 Minutes

Q2: (a) Using only four-bit **adders**, construct an eight-bit adder. Each four-bit adder has two four-bit inputs and one five-bit output. Your eight-bit adder should have two eight-bit inputs and a one eight-bit output.



(b) Using two 2-to-4 **decoders** with enable (EN) inputs, construct a 3-to-8 decoder. Provide a labeled logic diagram and explain how the enable signals are utilized.



(c) Implement the given Boolean function using an **8-to-1 multiplexer (8x1 MUX)**. Assume A_0 as the least significant bit (LSB).

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$$F(A_3, A_2, A_1, A_0) = \sum m(0, 3, 4, 5, 7, 11, 13, 14)$$

MUX Configuration

$$I_0 = A_0'$$

$$I_1 = A_0$$

$$I_2 = 1$$

$$I_3 = A_0$$

$$I_4 = 0$$

$$I_5 = A_0$$

$$I_6 = A_0$$

$$I_7 = A_0'$$

- (d) A 7-segment **decoder/driver** drives the display in the given Figure 1. If the waveforms are applied as indicated, determine the sequence of digits that appears on the display. [Solve here on the question paper].

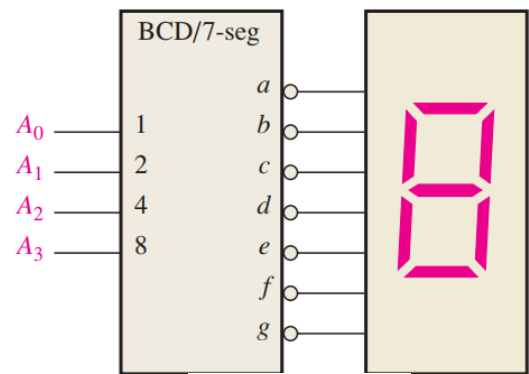
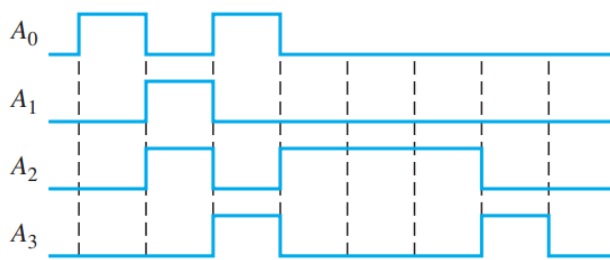


Figure 1

Start 0, 1, 6, 9, 4, 4, 4, 8, 0

- (e) The serial data-input waveform (Data in) and data-select inputs (S_0 and S_1) are shown in Figure 2. Determine the data-output waveforms on D_0 through D_3 for the **demultiplexer** in Figure 2. [Solve here on the question paper].

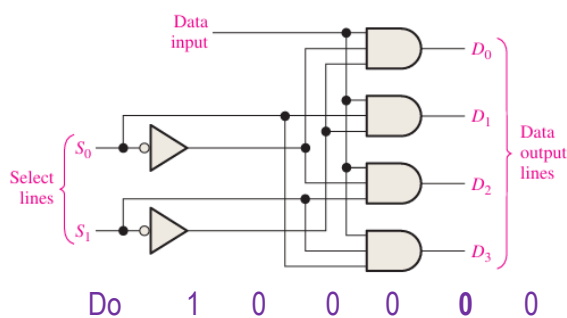
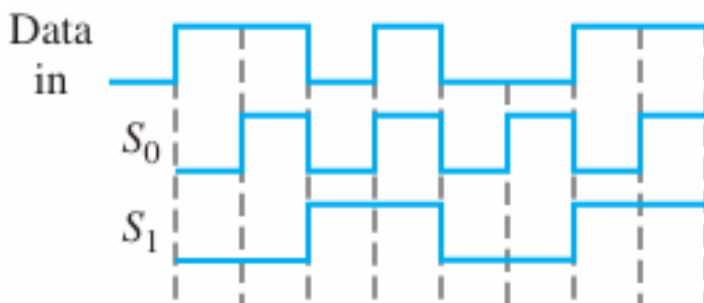


Figure 2

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D1	0	1	0	0	0	0	0	0
D2	0	0	1	0	0	0	1	0
D3	0	0	1	1	0	0	0	1

- (f) A **4-to-2 encoder** is a combinational circuit that converts one active input out of four into a 2-bit binary code. Now consider a **4-to-2 encoder** with inputs D_3, D_2, D_1, D_0 and outputs Y_1, Y_0 . Assume that **only one input is HIGH at a time**.

(i) Construct the **truth table** of the encoder.

D_3	D_2	D_1	D_0	Y_1	Y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

(ii) Derive the **Boolean expressions** for outputs Y_1 and Y_0 .

$$Y_1 = D_2 + D_3$$

$$Y_0 = D_1 + D_3$$

- (iii) What will happen if **more than one input is HIGH simultaneously**? Briefly explain.
If more than one input is HIGH at the same time:

- The output becomes **ambiguous (incorrect)**
- Example:
 - If $D_2=1$ and $D_3=1$
 - Output becomes:

$$Y_1=1, Y_0=1$$

- This corresponds to D_3 , but D_2 is also active → **information loss**

A simple encoder **cannot handle multiple active inputs**.
This limitation is resolved using a **priority encoder**.